

Performance Comparison of Session-based Recommendation Algorithms based on GNNs

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Introduction and Motivation

Recommender systems

Conventional recommender systems

Session based recommender systems
(SBRs)

❑ From a technical point of view

1. Most SBRs are based on neural models
2. Need user-item interaction data to generate a recommendation list

❑ Quite surprisingly

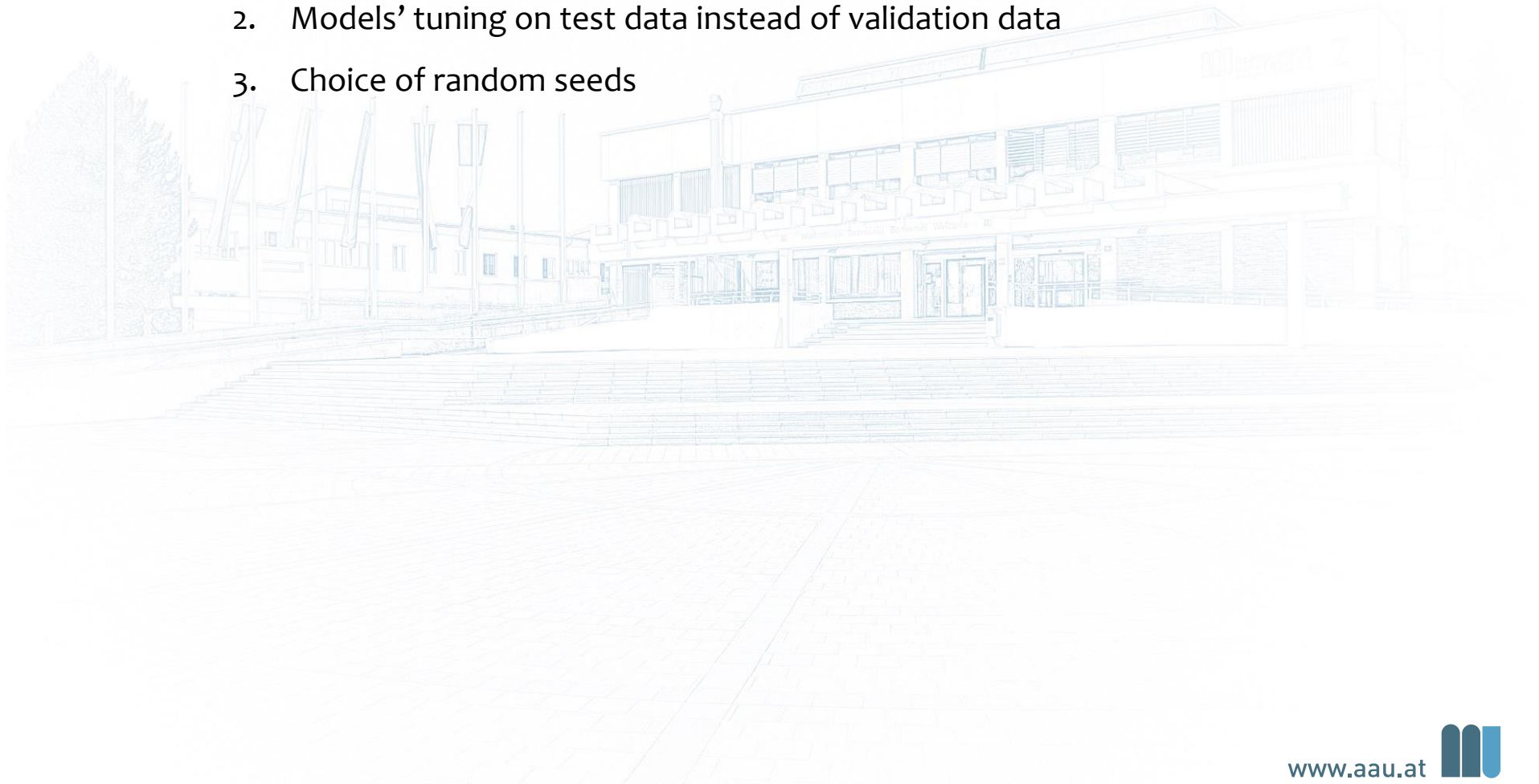
1. Many studies [1, 2] have shown that simple models (kNN, etc.) are competitive
2. Various factors [3] contribute to this “Phantom progress” such as compared to one family of algorithms (neural models), not properly tune baseline models, etc.

❑ An emerging trends in SBRs (graph neural networks)

Introduction and Motivation

❑ While examining the literature, we encounter some bad research practices

1. Fix the embedding size of models
2. Models' tuning on test data instead of validation data
3. Choice of random seeds



Research Question

- ❑ Although we know that simple models outperform the complex models in majority of cases. However, we want to know to what extent this is true for GNN models
- ❑ Could bad research practices lead to “virtual progress”?
 1. Fixing the embedding size of DL models
 2. Effect of random seeds
 3. Models’ tuning on test data
 4. Comparison with non-tuned models

Research Methodology

- ❑ Session-rec is a python based framework that is developed for conducting experiments for SBRS

Algorithm selection

- ❑ Graph neural networks
 1. Publication outlets: browsed top-outlets and identified 34 articles
 2. Reproducibility: only considered those articles for reproducibility for which code was available
 3. Eight articles [4-11] were reproduced and added them to the session_rec framework
- ❑ Baselines: select four strong baselines, which outperform the neural models in previous experiments

Research Methodology

❑ Dataset selection

1. Consider three datasets, which are mostly used in the identified articles and have category information used by two GNN models
2. RSC15 (RecSys Challenge 2015)
3. DIGI (Diginetica CIKM Cup 2016)
4. RETAIL (Retailrocket)

❑ Accuracy measures

1. Mean reciprocal rank (MRR)
2. Hit rate (HR)

❑ Beyond accuracy measures

1. Popularity bias
2. Training time (T-time)
3. Prediction time (P-time) requires to generate a recommendation list

Results

❑ RQ1: to what extent simple models outperform GNN models

Accuracy measures

Table 1: Best performing model for each dataset and each accuracy measure				
Datasets	RSC15 (1/64)	RSC15 (1/12)	RETAIL	DIGI
MRR@10	STAN	SR	VSTAN	SFSKNN
MRR@20	STAN	SR	VSTAN	SFSKNN
HR@10	CM-HGNN	VSTAN	VSTAN	SR-GNN
HR@20	CM-HGNN	GNRRW	VSTAN	SR-GNN

Note: GNN models are highlighted in blue color

❑ Findings

1. Simple models outperform all GNN models in case of MRR values on all datasets (MRR@20 was our optimization criterion)
2. In six cases, GNN models achieve the best HR@K
3. VSTAN and STAN consistently give competitive results for all datasets
4. More than a half decade, there have been strong baselines with more than 100 citations. However, no one considers them. why?
5. Empirical studies show the effectiveness of simple models (offline settings)

Results

❑ RQ1: to what extent simple models outperform GNN models

Additional quality measures

Table 2: Results are based on the highest values of Pop@20, T-time				
Datasets	RSC15 (1/64)	RSC15 (1/12)	RETAIL	DIGI
Pop@20	TAGNN	GCE-GNN	TAGNN	TAGNN
T-time (days)	GCE-GNN (0.16)	GCE-GNN (5)	MGS (1)	GCE-GNN (0.50)
Results are based on the lowest values of P-time				
P-time (ms)	SR (0.20)	TAGNN (0.30)	SR (0.40)	SFKNN (0.40)

Note: GNN models are highlighted in blue color

❑ Findings

1. kNN models recommend popular stuff, however, the results show a different behavior
2. T-time of GNN models depend upon datasets, number of items, session length, etc. (sometimes, they require days)
3. P-time of simple models lies near to the fastest GNN model and some GNN models require P-time more than 2 seconds, which is prohibitive in practice

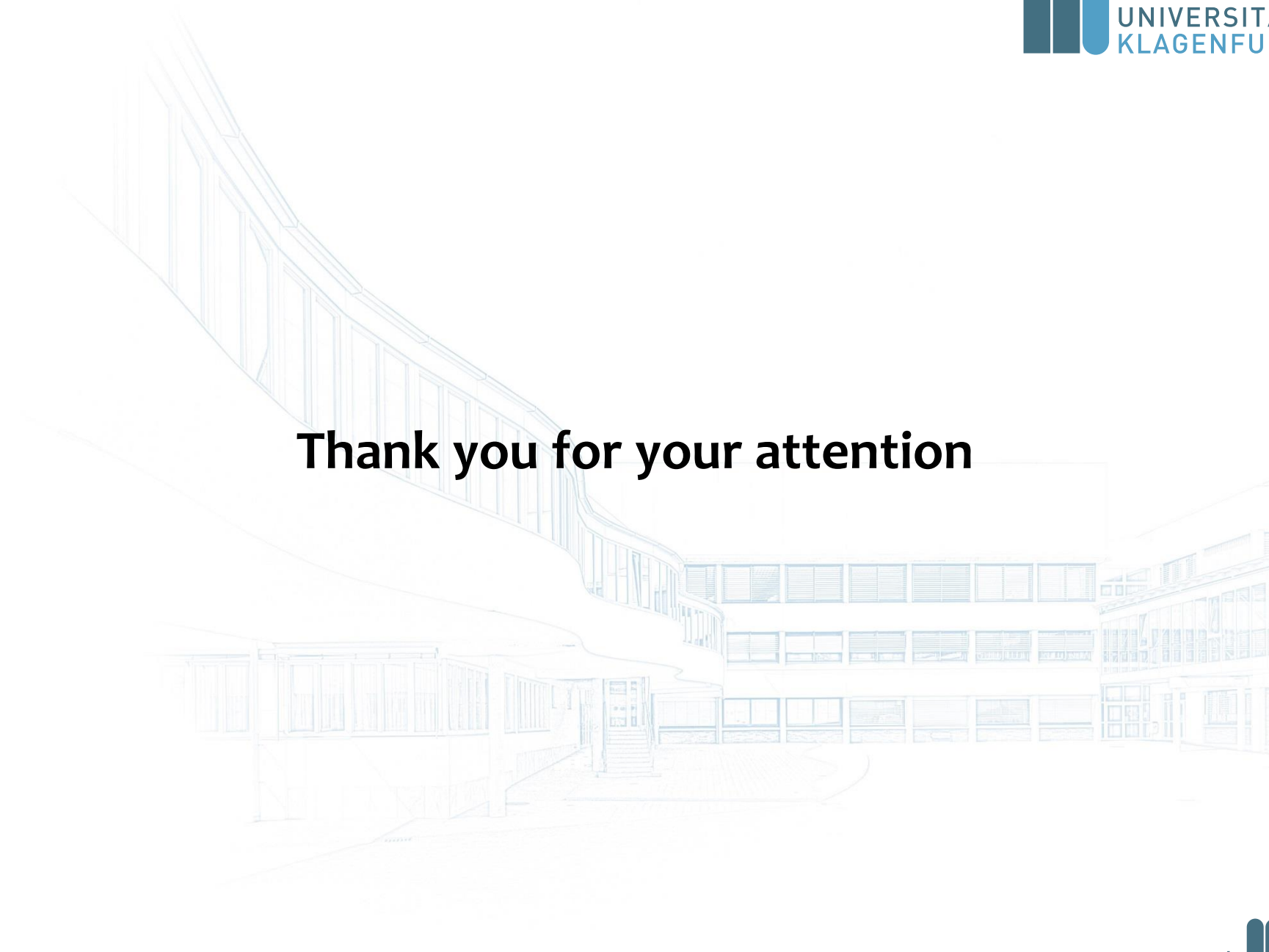
Results

❑ RQ2: Could bad research practices lead to “virtual progress”

Sr. No.	Bad Research Practices	Performance difference between the worst and the best MRR value of a model
1	Embedding size	> 40%
2	Random seeds	> 30%
3	Models' tuning on test data	3% to 4%
4	Comparison with non-tuned models	Everyone's a Winner! On Hyperparameter Tuning [12]

Conclusions

- ❑ Analysis of identified articles shows that new proposals were compared to
 1. With only recent algorithms
 2. One family of algorithms (neural models)
 3. Were not compared to simpler models that show competitive performance in empirical studies
- ❑ On a positive note, there is much room for improvement in the SBRS as there are not enough studies that take into account additional information of items. So, there is “still a long way to go”



Thank you for your attention

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